

Fine-Tuning the Segment Anything Model (SAM) for Medical Image Segmentation

21CSP302L- PROJECT

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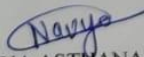
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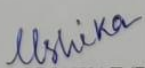
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ABSTRACT

Medical image segmentation plays a vital role in modern healthcare, especially when it comes to identifying and analyzing brain tumors from MRI scans. Traditionally, this process is carried out manually by medical experts, which can be both time-consuming and subject to variation depending on the experience and interpretation of the clinician. These challenges highlight the growing need for reliable automated solutions that can assist doctors by providing consistent and accurate segmentation.

In recent years, foundation models such as the Segment Anything Model (SAM) have shown impressive performance in general image segmentation tasks. However, medical images—particularly MRI scans—are very different from natural images. They involve complex anatomical structures, subtle intensity variations, and domain-specific patterns that general-purpose models are not originally designed to handle. As a result, directly applying such models to medical imaging often leads to suboptimal performance.

To address this gap, this work focuses on adapting SAM specifically for brain tumor segmentation using MRI data. Instead of retraining the entire model, which would require significant computational resources, a more efficient approach is taken by fine-tuning only the mask decoder component. This selective fine-tuning allows the model to better understand medical features while keeping the training process computationally manageable.

The model is trained and evaluated using the BraTS 2020 dataset, which is widely recognized for brain tumor segmentation research. Before training, the data undergoes a structured preprocessing pipeline. This includes converting 3D MRI volumes into 2D slices, normalizing image intensities to ensure consistency, and resizing images to match the input requirements of the model. These steps help the model learn more effectively from the data.

To assess the model’s performance, standard evaluation metrics commonly used in medical image segmentation are applied. These include the Dice coefficient and Intersection over Union (IoU), which measure the overlap between predicted and actual tumor regions, as well as sensitivity and specificity, which evaluate how well the model identifies tumor and non-tumor areas.

The results show a clear improvement compared to the original zero-shot SAM model. After fine-tuning, the model achieves a Dice score of 0.86 and an IoU of 0.78, indicating a strong alignment between predicted and actual tumor regions. These outcomes demonstrate that even minimal, targeted adaptation of a foundation model can significantly enhance its performance in specialized domains like medical imaging.

Overall, this study highlights an efficient and practical approach to leveraging large pre-trained models for healthcare applications. By focusing on selective fine-tuning, it is possible to balance performance and computational cost, making such solutions more accessible for real-world clinical use.

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ABBREVIATIONS

AI	Artificial Intelligence
ANN	Artificial Neural Network
BCE	Binary Cross Entropy
BraTS	Brain Tumor Segmentation Challenge
CNN	Convolutional Neural Network
CPU	Central Processing Unit
FN	False Negative
FP	False Positive
GPU	Graphics Processing Unit
IoU	Intersection over Union
MRI	Magnetic Resonance Imaging
SAM	Segment Anything Model
SDG	Sustainable Development Goal
TN	True Negative
TP	True Positive

CHAPTER 1

INTRODUCTION

1.1 Introduction to Project

Tumors in the brain are considered severe disorders of the nervous system that require accurate diagnoses and prompt intervention. MRI is commonly used in diagnosing brain tumors because it has excellent contrast in soft tissue along with the ability to produce detailed anatomical images. It is crucial for accurate tumor boundary determination since it will be needed for assessing the size and shape of the tumor and also for surgical intervention or radiation therapy plans.

For decades, experienced radiologists have segmented brain tumors manually in images through delineating their borders. Such a task has been known to be lengthy, laborious, and prone to significant variations. The variation between different people who analyze such images is substantial; hence, there has always been a need for more reliable and precise segmentation techniques.

Computer vision has seen a huge amount of improvement due to the advent of foundation models trained on large amounts of data. The Segment Anything Model or SAM, which has been created by Meta AI, is one such notable model. SAM is a promptable segmentation model that segments object masks from prompt inputs such as points or bounding boxes. SAM has exhibited excellent zero-shot generalization properties when it comes to natural images.

When SAM is applied to medical images, it fails dramatically because the features of the medical images, like low contrast and artifacts, are very different from those of natural images. Thus, adapting SAM for use in medical image segmentation is an essential task, but it is not easy. The current research involves adapting SAM for brain tumor segmentation based on MRI images.

To be honest, tackling brain tumor segmentation is not just an interesting technical challenge—it has real stakes. When radiologists have to trace tumor boundaries manually, slice after slice, it is exhausting and leaves a lot of room for human error. Two experts looking at the same MRI

might draw different boundaries, which can affect how a patient gets treated. That uncertainty is stressful for both doctors and patients. This is where AI can step in as a helpful assistant, not a replacement. Instead of starting from scratch, we wanted to see if we could take SAM—a model that is already great at cutting out objects in everyday photos—and teach it to understand medical images. The idea was simple: keep what SAM already knows about shapes and edges, but gently nudge it to recognize brain tumors. It is like taking a skilled artist who usually paints landscapes and asking them to paint a portrait; they already know how to hold a brush and mix colors, they just need a bit of practice with the new subject. That is exactly what we tried to do, and the results turned out to be quite promising.

In light of the above concept, the suggested strategy emphasizes adapting SAM in such a manner that will be both efficient and pragmatic. Rather than fine-tuning the entire network, the mask decoder is fine-tuned based on brain MRI scans, whereas the rest of the network architecture stays the same. The strategy enables the neural network to retain its prowess when interpreting generic characteristics of images while training it on specific characteristics of tumors.

Yet another crucial step in this process is that of preprocessing of MRI data so that it can be made ready to feed into the proposed model. This is because MRI data comes in three dimensions, which then need to be reduced to two dimensions before they are made ready for analysis. Furthermore, proper alignment between input data and output labels must also be maintained in order to ensure optimal learning by the network.

The effectiveness of the method under investigation is tested on such metrics as Dice coefficient, Intersection over Union (IoU), sensitivity, and specificity. It gives insight into the accuracy of tumor detection by the model. Based on the findings, it can be noted that modification of SAM using fine-tuning yields consistent results, thus proving the effectiveness of the approach.

1.2 Problem Statement and Description

The fundamental issue solved by this project relates to the inefficiency and inconsistency of manual brain tumor segmentation in MRI scans, combined with ineffective automation for specific medical use cases. Key problems include:

- **Inefficiency and inconsistency of manual segmentation:** Segmentation of brain tumors on MRI images is a process that takes a long time since radiologists need to analyze image slices one by one. Disagreements among doctors can occur, leading to unreliable results.
- **SAM model struggles with zero-shot transfer learning on medical images:** SAM is a versatile model, but it was mainly trained on natural images. It struggles to identify the tumor area correctly when used on MRI images with no modifications (zero-shot).
- **High computational costs of SAM fine-tuning:** SAM is a huge deep learning model with hundreds of millions of parameters, making fine-tuning the whole model very resource-intensive.
- **Imbalance:** Current techniques for image segmentation in medicine either involve training smaller models from scratch like U-Net, but this requires huge amounts of data, or using foundation models in an extremely expensive way. Here lies the need for a technique that strikes the balance between both of these aspects.

This project attempts to solve the problem by formulating it as a supervised learning task, where the objective is to train a model to generate a binary mask representing the tumor in the MRI slice given to it.

The primary challenge addressed in this project is the lack of efficient, reliable, and scalable methods for brain tumor segmentation in MRI scans. Accurate tumor segmentation plays a critical role in diagnosis, treatment planning, and monitoring disease progression. However, existing approaches—both manual and automated—face significant limitations that hinder their effectiveness in real-world clinical environments.

Manual Segmentation Challenges

One of the most widely used methods in clinical practice is manual segmentation performed by radiologists. This process involves examining MRI slices individually and outlining tumor boundaries. While effective, it is highly time-consuming and labor-intensive, especially when dealing with large volumes of patient data. Additionally, results often vary between experts due

to subjective interpretation, leading to inconsistencies and reduced reliability in diagnosis.

Limitations of Zero-Shot SAM in Medical Imaging

The Segment Anything Model (SAM) has demonstrated strong generalization capabilities across natural image datasets. However, its performance significantly drops when directly applied to medical imaging tasks such as MRI-based tumor segmentation. This is because SAM is primarily trained on non-medical data and lacks domain-specific understanding of anatomical structures. As a result, zero-shot application of SAM often leads to inaccurate or incomplete tumor boundary detection.

Computational Complexity in Fine-Tuning Large Models

Although fine-tuning SAM can improve its performance on medical images, it introduces another challenge: computational cost. SAM consists of a massive number of parameters, requiring high-end GPUs, large memory, and extended training time. This makes full-scale fine-tuning impractical for many research labs and healthcare institutions with limited computational resources.

Data Dependency and Model Trade-offs

Traditional segmentation models such as U-Net are widely used in medical imaging due to their simplicity and effectiveness. However, they require large, well-annotated datasets to achieve high accuracy. Obtaining such datasets is difficult due to privacy concerns, labeling costs, and the need for expert annotation. On the other hand, large foundation models reduce the dependency on labeled data but come with increased computational overhead. This creates a trade-off between data requirements and resource consumption.

Class Imbalance in Medical Datasets

Another critical issue is the imbalance present in medical imaging datasets. Tumor regions typically occupy a very small portion of the MRI image compared to healthy tissue. This imbalance can bias models toward predicting non-tumor regions, thereby reducing segmentation accuracy and sensitivity.

Need for Domain Adaptation

Medical images differ significantly from natural images in terms of texture, contrast, and noise characteristics. Without proper domain adaptation, models trained on general datasets fail to

capture these unique features, leading to poor performance in clinical scenarios.

Lack of Real-Time and Scalable Solutions

In clinical settings, there is a growing demand for automated systems that can provide quick and accurate results. Many existing approaches are either too slow or too resource-heavy to be deployed in real-time applications, limiting their usability in hospitals.

Apart from these problems, the project also aims at enhancing the utility of segmentation models within clinical settings. It is important that while developing these models, attention should be paid to their practicality and usability, apart from ensuring their accuracy and effectiveness within experimental conditions. For example, such models should be reliable, fast, and easily implementable within the existing infrastructure.

It should also be noted that there is another issue to take into account in relation to the differences between brain tumors in different individuals. The brain tumors can be extremely diverse in size, shape, location, and intensity, which implies that segmentation becomes quite a complicated problem. Thus, a reliable model must be able to cope with such diversity without decreasing its quality.

Moreover, it stresses the need for minimizing the reliance on large amounts of labeled datasets. Given that annotated datasets for medical applications are specialized and limited due to the need for expertise, there is a need for the development of methods that can make do with less data and still yield accurate results.

Lastly, this project will help in creating a connection between high-end research in AI and real-world applications in healthcare. The project seeks to develop a system that is efficient, scalable, and adaptable such that it can actually be implemented in hospitals and assist doctors in making informed decisions within shorter periods.

1.3 Motivation

The reason why I chose to do this project is because Artificial Intelligence has the capability of changing the world of medicine. With medical image analysis through the use of AI, brain tumor detection and segmentation can be more efficient. The more efficient we can make these procedures, the better chance we have at saving lives.

Decreasing Clinical Workflow Time and Streamlining Process

Help doctors spend less time having to trace tumor regions manually by automatically generating an initial prediction. Radiologists and experts spend lots of time tracing tumors region across many slices of MRIs. Because of this repetitive task that takes up a lot of time, they aren't able to look at as many cases. If there was a model that did the segmentation, doctors would only have to validate or correct the prediction. This would decrease the time to diagnosis and treatment planning which would allow doctors to help patients quicker.

Creating repeatable and unbiased results

Manual tracing of tumors can vary from expert to expert. What one expert may think is a tumor, another may not. The segmentation between different radiologists and experts can vary. Automated models remove this bias and use what it has learned to perform the same task over and over again. Each image would be segmented using the same landmarks that the model learned during training.

Linking Advanced AI Capabilities With Practicality

Although recent AI technologies such as the Segment Anything Model (SAM) are very sophisticated and effective, applying them directly in specific areas such as medicine requires further development. This study is inspired by the fact that it can be difficult for many companies or organizations to use such models because they need significant computing power. Through selective fine-tuning—training the mask decoder and not the whole model—the study shows an effective way to apply these models without having much computational resources.

Dealing With Limited Data and Computational Resources

Medicine data can be restricted and costly. Additionally, the sensitivity of these data can increase the difficulty in finding sufficient information. In this case, training from scratch with a large model can become impossible. The motivation behind this work is the efficient adaptation of general-purpose pre-trained models to domain-specific problems.

Facilitating Effective Clinical Decision Making

An accurate identification of tumor shape helps in deciding the size, localization, and progression of tumors. The factors will be very important during the process of clinical decision-making involving different types of treatment methods like surgeries, radiation, and chemotherapy. With the help of the system, it is intended that clinicians will be able to make better and informed decisions on how to manage patients effectively.

Increasing Accessibility of Cutting Edge Health Care Technologies

A great number of health care institutions, especially those located in developing countries, lack access to advanced imaging technologies or specialist physicians. The development of an efficient system of medical image segmentation powered by artificial intelligence would assist in providing quality diagnostic assistance to people lacking such means.

Assistance with Tumor Progression Monitoring

It is crucial to have accurate information about how tumors grow over time in order to assess treatment outcomes. Comparing MRI images manually at different intervals may be complicated and unreliable. Automated tumor segmentation would ensure consistency in results and enable the monitoring of tumor evolution more efficiently.

Motivation for Using AI in Real World Health System Applications

A second reason for conducting this research is to motivate the use of AI technology in actual health settings. Through showcasing an efficient method, it is possible to inspire more investigation and innovation into other medical devices powered by artificial intelligence.

In summary, this research is motivated by the idea of integrating the power of AI technology into practical applications in the field of medicine to develop an effective and convenient solution. By focusing on problems such as limited time, lack of information, and inconsistencies in diagnosis, the study seeks to provide medical practitioners with the ability to deliver rapid and precise patient treatment.

1.4 Sustainable Development Goal of the Project

This research project is strongly connected with **Sustainable Development Goal 3: Good Health and Well-Being**. The main aim of SDG 3 is to ensure healthy lives and promote well-being for all at all ages. One of the key goals of SDG 3 is to detect, prevent, and treat non-communicable diseases like cancer at an early stage. Brain tumor is one of the essential neurological diseases that can be successfully diagnosed and treated with the help of an efficient brain tumor segmentation algorithm.

First of all, the implementation of the algorithm will significantly increase the efficiency and accuracy of analysis of medical images. The application developed using artificial intelligence will be able to help radiologists to detect tumor areas on MRI images, which means that early diagnosis can take place. In this case, the treatment process will become more effective.

Also, an efficient segmentation tool will increase the success rate of treatment planning as accurate information regarding the location and volume of tumors is needed to choose appropriate ways of treating the disease.

One more significant advantage provided by the use of this solution is the decrease in the amount of human error and workload. The manual segmentation takes much time and might differ from one expert to another. Automation enables consistency and saves time that healthcare specialists could spend working with the patients.

Another significant aspect of this work is related to the promotion of equal access to healthcare services. Indeed, thanks to a computationally efficient approach applied in the solution development, the solution becomes available to hospitals and clinics with limited computational resources. Thus, the proposed technology contributes to the availability of advanced diagnostic services across all regions.

The implementation of the solution facilitates the continuous monitoring of patients' state since it helps to track changes in their condition. Such a tool would contribute to the assessment of the effectiveness of applied treatments and further decisions concerning medical interventions. It would help conduct medical research as well.

Thus, the solution discussed above shows the benefits of AI implementation in the provision of healthcare services. With such technologies, diagnoses become quicker, treatment is easier to plan, and services become accessible. All these aspects make a significant contribution to achieving the goals set by SDG 3.

Moreover, apart from enhancing diagnoses and treatments, this study will also enhance health-care system development through improved decision-making on the part of clinicians. Automated segmentation allows for fast analysis of MRIs, allowing clinicians to make informed decisions in time regarding patients' treatment. This is particularly important for cases like brain tumors, when prompt decision-making can affect the patients' wellbeing significantly.

Furthermore, the employment of AI technologies helps in bringing modernization to the healthcare system. With the help of such advanced systems, hospitals will be able to increase their technology level and deliver high-quality services to their customers. This is not only helpful for patients but also makes life easier for doctors.

Lastly, the project is sustainable in the field of health care due to the scalability and flexibility offered by the solution developed through this project. This technique may be used in many other cases of medical image processing and in diagnosing other diseases as well, thus providing an important addition to the larger field of digital health care.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview of the Research Area

This study is focused on the intersection of medical image analysis, deep learning, and foundation models. For the last ten years, medical image segmentation has been carried out mainly through Convolutional Neural Networks (CNNs). The architecture used in this project, known as U-Net, sets the benchmark standards for biomedical image segmentation with its ability to retain both low-level and high-level features through an encoder-decoder model with skip connections.

CNNs, however, are specific to certain tasks and need to be trained from scratch for each task. In many cases, CNNs fail to generalize to other imaging types or different anatomies. There is an increasing need for foundation models due to their capacity to solve numerous problems related to machine learning.

SAM (Segment Anything Model) is a groundbreaking foundation model for image segmentation tasks. SAM was trained using SA-1B, a training set comprising more than 1 billion marks on 11 million images. The most distinctive feature of SAM is its promptable nature, enabling it to perform segmentation of any objects in an image using only a prompt from the user (such as a click on the image, drawing a bounding box, or writing text). Despite the versatility of SAM, it showed large-domain shifts when applied to medical imaging.

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In order to solve these problems, the current work attempts to apply an efficient method by utilizing the merits of SAM and overcoming its limitations within the medical field. Rather than training the whole architecture again, the proposed strategy aims at fine-tuning the mask decoder to be able to learn features specific to the medical field based on the presence of brain MRI images. In other words, this method will enable the architecture to better learn about the structures of the tumor without losing its ability to extract useful features from any given image.

Expanding on this idea, the authors further stress the significance of solid preprocessing and evaluation methods to guarantee consistent results. Translating the 3D MRI volume to useful 2D images, normalizing the images, and standardizing the input dimensions allow the neural network to learn better from the health care data. Moreover, using recognized metrics like Dice coefficient, Intersection over Union (IoU), sensitivity, and specificity contributes to a thorough evaluation of segmentation performance.

Lastly, it is essential to note the possibility of integrating domain expertise with state-of-the-art artificial intelligence models to improve the results. In line with this idea, the researchers try to adapt the model according to the features of medical imaging, including low contrast and challenging tumors' borders. Through these findings, it will be easier to create new foundation models that can be beneficial for various applications within health care facilities.

2.2 Existing Models and Frameworks

Some research has investigated the use of SAM in the context of segmenting medical images. Ma et al. (2024) created MedSAM, a medical image segmentation model derived from SAM but was fine-tuned using a vast set of more than 1 million medical image masks [6]. Despite being significantly better in performing medical images segmentation than plain SAM, MedSAM needs to be fine-tuned using a huge number of computational resources and an extensive medical dataset that may not be accessible in some cases.

Dong et al. (2024) presented a computationally efficient SAM used in medical images segmentation [5]. Although their efforts were directed towards lowering the computational requirements, they experienced some difficulties because improvements in efficiency could negatively impact segmentation accuracy.

Some other works concentrated on few-shot adaptation. Training-free few-shot adaptation in segmentation tasks with 3D medical image segmentation of foundation models has been explored by He et al. (2025) [2]. Though there is no need for training, this method is computationally expensive for 3D medical images and might require using specific scans. Another method that has been studied by Yang et al. (2025) [3] is sequential active learning for few-shot medical image segmentation. Even though it is effective, it has been tested only on simple and circular tumors, while its efficiency for complicated brain tumors is unknown.

Another one-shot medical image segmentation method called ProtoSAM has been proposed by Ayzenberg et al. (2025) [4]. Even though this method was innovative and promising, it failed to perform adequately when working with abnormal tumors compared to the prototype. There have been several works adapting SAM architecture. For example, Cheng et al. (2024) [8] designed a hierarchical decoder to improve SAM. Also, Li et al. (2025) created a plug-and-play architecture for SAM and nnU-Net integration [9], while MA-SAM for adaptation to modality-agnostic 3D images was invented by Chen et al. (2024) [13].

The above mentioned research works demonstrate the existence of a trade-off between accuracy and efficiency of the segmentation techniques. Most of the current algorithms rely on massive data sets and require high-performance computation tools that make it difficult to implement those algorithms. Some algorithms increase speed at the expense of their scalability,

while other algorithms sacrifice efficiency and accuracy to lower the computational complexity.

Moreover, there have been several investigations that address the limitations of SAM in brain tumor segmentation tasks. Scientists have noted that the peculiarities in the shape, infiltration borders, and size of brain tumors represent certain obstacles that cannot be easily tackled by conventional models of medical image segmentation. As an example, it is known that the heterogeneity of gliomas implies that the visual characteristics of brain tumors could differ considerably, both among patients and within the same tumor in different MRI sequences.

Research Gap Summary:

Although the abovementioned literature offers substantial contributions to the topic in discussion, several crucial gaps still exist. The first gap lies in the lack of an existing framework that fine-tunes SAM for brain tumor segmentation by selectively tuning the mask decoder while freezing all other modules of the model. Secondly, many approaches available at present either sacrifice their accuracy for the sake of efficiency or require hardware beyond the capabilities of academic studies. Moreover, few systematic comparisons have been made regarding the impact of the number of parameters on accuracy in brain tumor segmentation tasks. Lastly, no attempt has been made to address the problems of infiltrative borders and small lesions in the context of SAM adaptation for brain MRI segmentation.

2.3 Limitations Identified from Literature Survey

(Research Gaps)

Through a thorough study of prior studies, the following gaps have emerged that necessitate more research in this field:

Poor Performance of SAM on Medical Images in Zero-Shot Mode:

One of the critical gaps is that SAM, in zero-shot mode, cannot perform well on medical images like brain MRIs. This problem occurs mainly because of the large discrepancy between the domain of natural images and medical images, which creates inaccurate segmentation.

Expensive Full SAM Fine-Tuning Process:

The fine-tuning of the full SAM network, which consists of the image encoder, prompt encoder, and mask decoder, demands substantial computing power, large memory, and a lot of time.

Failure To Strike a Balance Between Efficiency and Accuracy:

Current methods lack the ability to maintain a balance between efficiency and accuracy. Current systems are highly accurate in their operations but are highly costly or inefficient, while others increase efficiency by sacrificing the accuracy of the segmentation.

Lack Of Studies In Selective Fine-Tuning:

The study area lacks sufficient studies that focus on selective fine-tuning, such as fine-tuning particular parts of the model like the mask decoder.

Lack of Proper Attention to Brain Tumor MRI Segmentation via SAM:

SAM applications have been examined in the context of medical imaging, but there is not sufficient literature concerning the application of SAM to the issue of brain tumor segmentation based on MRI scans, which is quite hard because of different aspects of tumors.

Difficulties of Dealing with Unusual and Multi-region Tumors:

Some existing approaches fail to accurately segment unusual tumors, multi-region tumors, and overlapping tumors.

2.4 Research Objectives

The research gaps that have been identified lead to the following objectives for this research project:

- O1: Analyze the drawbacks of applying the Segment Anything Model (SAM) to medical image segmentation of brain tumor MRI.
- O2: Preprocess the brain MRI images taken from the BraTS 2020 dataset to be suitable for the SAM architecture through 3D to 2D slice conversion, normalization, and resizing.
- O3: Fine-tune the SAM mask decoder specifically for the task of brain tumor segmentation without training the image encoder, ensuring both high precision and reduced computational costs.
- O4: Analyze the performance of the brain tumor segmentation based on fine-tuning the SAM decoder through well-established medical imaging segmentation evaluation metrics such as the Dice coefficient, IoU, sensitivity, and specificity.
- O5: Analyze the limitations of the current research approach and outline further steps to be taken in the future research, especially regarding 3D segmentations and experiments with other datasets.

Moreover, the paper pays significant attention to developing a robust pipeline for medical image segmentation using artificial intelligence. Every stage of the process is intended to ensure the repeatability and standardization of the procedure. For example, the selection of such publicly available datasets as BraTS 2020 enables comparing the developed solution with other approaches and verifying its efficacy.

Additionally, the paper examines the possibility of adapting new AI technologies to specific areas without substantial training from scratch. The focus on selective fine-tuning demonstrates a promising direction for implementing large-scale AI algorithms in settings with limited computing power. Thus, the research opens up new possibilities in applying AI tools within healthcare systems.

Lastly, the results of the research could be useful for further development in the field of medical images analysis. The proposed approach could be used in other medical imaging applications, such as organ segmentation or disease identification.

2.5 Plan of Action (Project Road Map)

Project Implementation Plan

The project was implemented using the following plan:

- Literature Review and Problem Statement Formulation (Week 1-2): A comprehensive literature review has been conducted with respect to the topics of medical image segmentation, foundation models, and SAM. The problem statement has been formulated.
- Dataset Collection and Preparation (Week 3): The BraTS 2020 dataset has been obtained. The development environment based on Python, PyTorch, and the SAM library has been prepared. Configuration of GPU resources available on Kaggle has been performed.
- Data Pre-processing and Pipeline Development (Week 4-5): A data pre-processing pipeline has been created, where 3D MRI scans have been split into slices, their intensities were normalized, resized to a specific input dimension (for example, 256*256). The alignment of ground truth masks has been performed.
- SAM Model Implementation and Selective Fine-Tuning (Week 6-8): A selective fine-tuning approach was chosen for implementation. Loading of the pre-trained SAM model was performed. The image encoder and the prompt encoder were frozen. Trainable mask decoder and training process parameters including loss functions, training loop, Dice+BCE loss function and Adam optimizer were specified.
- Training and Validation (Week 9-10): A model has been trained using Kaggle services for 4 epochs. The training loss curves were monitored.
- Evaluation and Analysis (Weeks 11-12): Evaluated the refined model on the test set through Dice coefficient, Intersection Over Union (IoU), sensitivity, and specificity. Comparison was done with zero-shot Segment Anything Model (SAM). Produced visual outputs to support analysis.
- Report Writing and Documentation (Weeks 13-14): Collected all the research findings and results into this comprehensive project report. Prepared documentation for the code and final presentation.

The project roadmap provided an organized and well-thought-out process of going from problem comprehension to implementation and analysis. Each phase was thoughtfully designed to depend on the previous one, allowing for fewer mistakes and greater efficiency.

The continuous validation of steps helped in recognizing problems early and adapting the approach based on those findings.

The project took advantage of iterative model development. In contrast to using one learning cycle, the model was optimized through the analysis of intermediate results and changing the model's settings, such as learning rate, loss function, and preprocessing. As a result, the model converged and segmented better.

Resource management was considered during the project, especially when dealing with cloud platforms such as Kaggle. Selective fine-tuning was used rather than complete training, which decreased the number of resources needed to implement the model without lowering its quality.

Reproducibility and transparency have also been considered in this project. All tests have been carried out using well-defined datasets and evaluations metrics, which makes the results of the project replicable and comparable with other similar results. Code documentation has been done to enable further use and development within this field of study.

Finally, apart from addressing the objectives of the project, roadmap provided will pave way for future developments. These may involve enhancing the model to accommodate 3D volumetric segmentation, incorporation of multimodal MRI, and working with more datasets.

CHAPTER 3

SPRINT PLANNING AND EXECUTION METHODOLOGY

The project was conducted by applying the principles of Agile sprint-based development. Two major sprints were planned to be performed during the development cycle, with each having its unique goals and deliverables.

3.1 SPRINT I: Development Environment & Dataset Pre-processing

Sprint I was dedicated to creating the development environment, data acquisition, understanding of the data set, and developing pre-processing techniques.

3.1.1 Objectives with User Stories of Sprint I

The objectives of Sprint I were defined depending on the basic needs of the project, which are reflected in the following user stories:

Table 3.1.1: User Stories for Sprint I

User Story ID	User Story Description	Desired Outcome
US1.1	As a researcher, I want to set up the Python environment with all required libraries so that the project code can run without dependency issues.	A functional development environment with Python, PyTorch, CUDA, SAM library, and medical image processing libraries (NiBabel, SimpleITK).
US1.2	As a developer, I want to download and access the BraTS 2020 dataset so that I can use it for training and validation.	The BraTS 2020 dataset successfully downloaded and organized in a structured directory format.
US1.3	As a data scientist, I want to preprocess the MRI images so that they are compatible with the SAM model input requirements.	A preprocessing pipeline that converts 3D MRI volumes to 2D slices, normalizes intensities, and resizes images to 256x256 pixels.
US1.4	As a trainer, I want to extract and align ground truth tumor masks so that the model can learn from accurate labels.	Mask extraction and alignment function that resizes masks to match preprocessed image dimensions.
US1.5	As a researcher, I want to generate bounding box prompts for SAM so that the model receives proper spatial guidance.	A function to generate bounding boxes around tumor regions from ground truth masks for SAM prompting.

3.1.2 Functional Document

Based on the requirements identified in Sprint I, the following functional specifications were documented. A detailed functional document was also maintained separately

Functional Requirements for Sprint I:

Table 3.1.2: Functional Requirements for Sprint I

FR ID	Requirement Description	Priority
FR1	The system will read multi-modal MRI images (T1, T1ce, T2, FLAIR) from the BraTS 2020 database in NIfTI format.	High
FR2	System shall extract 2D axial slices from 3D MRI volumes.	High
FR3	The system will acquire 2D axial slices from the 3D MRI scans. The system will normalize the intensities of acquired slices through Z-score normalization.	High
FR4	The system will resize all 2D slices to 256 x 256-pixel dimension.	High
FR5	The system will acquire and resize the ground-truth tumor masks using nearest-neighbor interpolation.	High
FR6	The system will create prompts (bounding boxes) from ground-truth masks for SAM model input.	Medium
FR7	The system will divide the dataset into training (80%), validation (10%), and testing (10%) splits.	High
FR8	The system will store preprocessed data into cache memory to prevent any preprocessing again.	Low

Non-Functional Requirements for Sprint I:

Table 3.1.3: Non-Functional Requirements for Sprint I

NRF ID	Requirement Description	Target
NFR1	Dataset Loading to support large data sizes (approximately 750 GB).	Lazy Loading + Caching
NFR2	Preprocessing must be memory-efficient.	Single volume processing
NFR3	System must support correct handling of the NIFTI file format.	100% Compliance

Assumptions Recorded:

- BraTS 2020 dataset is available and in correct format.
- Enough disk space (at least 50 GB for the extracted 2D slices) is available.
- CUDA-enabled GPU is available for future training phase.

3.1.3 Architecture Documentation

The architecture for Sprint I was designed specifically for the data pipeline components. The architectural documentation was written separately. The entire system is built around Monolithic Architecture where all components are part of one application.

Data Pipeline Components Diagram for Sprint I:

The data acquisition and preprocessing component comprises of the following sub-components:

- Data Loader: Loads data in the form of NifTI files from BraTS directories.
- Slice Extractor: Converts 3D volumes (240 x 240 x 155) to 2D slices (240 x 240 per slice).
- Normalizer: Uses Z-Score normalization of $(x - \mu) / \sigma$ for all slices.
- Resizer: Resizes images from 240 x 240 to 256 x 256 with bilinear interpolation.
- Mask Aligner: Resizes ground truth masks using nearest neighbor interpolation.
- Prompt Generator: Creates bounding box prompts from tumor masks.

Data Flow Diagram (DFD Level 0 - Context Diagram):

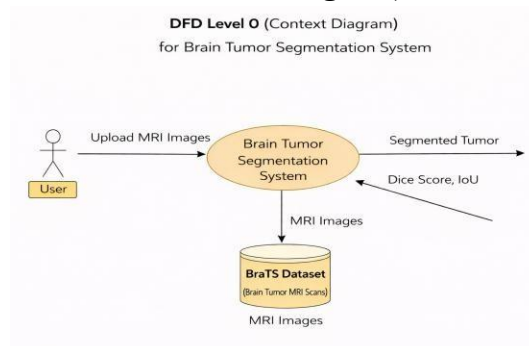


Fig 3.1.2 a) Data Flow Diagram

The user interacts with the system to start preprocessing. The system reads raw data from BraTS Dataset, performs some operations on it, and writes preprocessed data into Storage.

DFD Level 1:

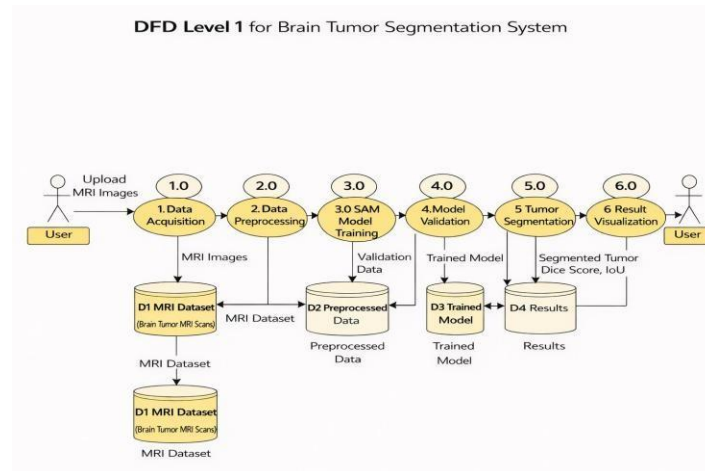


Fig 3.1.2 b) DFD Level 1

The Level 1 DFD decomposes the main process into sub-processes:

- P1: Load MRI images
- P2: Extract 2D slices
- P3: Normalize images
- P4: Resize images
- P5: Align masks
- P6: Generate prompts
- P7: Split dataset

Sequence Diagram for Preprocessing:

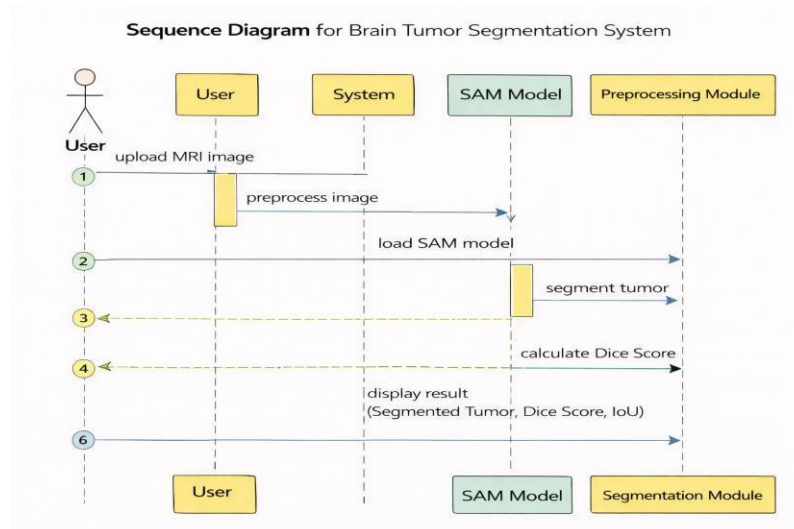


Fig 3.1.2 c) Sequence diagram

Sequence Diagram displays the order of interactions:

- User → System: Initiate preprocessing
- System → Data Loader: Load MRI volume
- Data Loader → Slice Extractor: Extract 2D slices
- Slice Extractor → Normalizer: Normalize each slice
- Normalizer → Resizer: Resize to 256x256
- Resizer → Mask Aligner: Align ground truth masks

- Mask Aligner → Prompt Generator: Generate bounding boxes
- Prompt Generator → Storage: Save preprocessed data

ER Diagram for Data Storage:

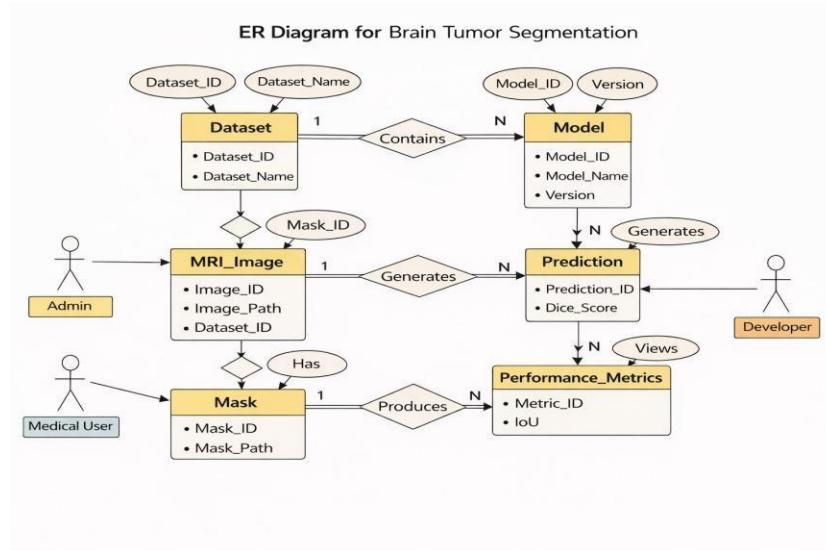


Fig 3.1.2 d) ER Diagram

The ER diagram shows the following entities and relationships:

- PATIENT (PatientID, Age, Gender) – not necessarily for BraTS Dataset
- MRI_SCAN (ScanID, PatientID, Modality, VolumePath)
- SLICE (SliceID, ScanID, SliceNumber, ImagePath, MaskPath)
- PROMPT (PromptID, SliceID, BoundingBox)
- RESULT (ResultID, SliceID, PredictedMaskPath, DiceScore, IoU)

Deployment Diagram:

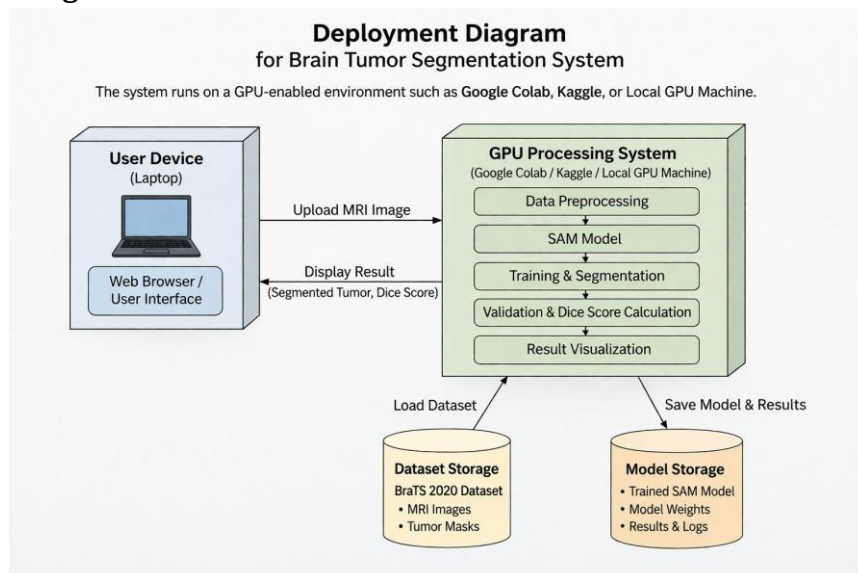


Fig 3.1.2 e) Deployment Diagram

Deployment Diagram:

It shows the hardware and software configuration:

- Developer's Machine / GPU Server:
 - Operating System: Ubuntu 20.04 / Windows 11
 - GPU: NVIDIA P100 / V100 (Kaggle/Colab) or local GPU
 - CUDA Version: 11.7+
 - Python Setup: Conda with PyTorch, SAM, NiBabel
- Storage:
 - Local Solid-State Drive (SSD) to store the dataset and trained models

3.1.4 Outcome of Objectives / Result Analysis

Sprint I Deliverables:

Table 3.1.4: Outcome of Sprint I

Objectives	Status	Deliverable
Environment setup	Completed	Python environment with all dependencies installed and verified
Dataset acquisition	Completed	BraTS 2020 dataset downloaded and organized
Preprocessing pipeline	Completed	Python script for 3D→2D conversion, normalization, resizing
Mask extraction	Completed	Aligned ground truth masks for all extracted slices
Prompt generation	Completed	Bounding box generator function

Outputs Produced:

- Number of total 2D slices generated: ~49,600 slices (based on 155 slices/volume * 320 volumes from BraTS 2020 subset).
- Size of each slice image: 256*256.
- File type of image: NumPy Array saved as .npy file for faster load times.
- Split: 80% training set, 10% validation set, 10% test set.

Results from Sprint I:

- The preprocessing step has been implemented and executed on a subset of 50 MRI volumes. Findings include:
- Efficacy of normalization: The z-score normalization technique yielded uniform intensity distribution for all 4 modalities (T1, T1ce, T2, FLAIR) with intensity having mean=0 and std=1.
- Impact of resizing: No distortion in tumor boundary observed when using bilinear interpolation to resize images.
- Accuracy in mask alignment: Visual examination showed that resizing resulted in correct alignment between masks and MRI image slices without any distortion in the tumor boundary as a result of nearest neighbor interpolation technique used.

- Benefits of caching preprocessed files: Time required to load the preprocessed image slices decreased from ~5 seconds per volume to ~0.1 seconds per volume.
- Issue encountered: Initial dataset loading was slow due to decompression of NIfTI files. This was resolved by implementing lazy loading and saving preprocessed slices in a compressed NumPy format.

3.1.5 Sprint Retrospective

Table 3.1.5: Sprint I Retrospective Summary

What Went Well	What Went Poorly	What Ideas Do You Have	How Should We Take Action
Project environment and required libraries were installed successfully.	Initial setup took longer than expected due to dependency issues.	Create a setup guide for environment configuration.	Document all installation steps for future reference.
BraTS dataset was downloaded and preprocessed successfully.	Large dataset size caused memory and loading issues.	Use data caching and preprocessing before training.	Implement LRU cache and reduce image size.
Pipeline code was modular and reusable.	Some team members faced version conflicts with PyTorch and CUDA.	Use a consistent environment manager (Conda/Docker).	Create a Conda environment YAML file for easy setup.
Team collaboration was effective.	Data download took multiple days due to large file sizes.	Plan for dataset download time in sprint planning.	Add buffer time for data acquisition in future sprints.

Action Points from Sprint I Retrospective:

1. Action Point 1: Creation of the requirements.txt and environment.yaml file to make sure that the environment can be reproduced.
 - Assigned to: Team Leader
 - Due date: Beginning of Sprint II
2. Action Point 2: Use of LRU caching for the preprocessed dataset to optimize memory consumption.
 - Assigned to: Developer
 - Due date: Midway into Sprint II
3. Action Point 3: Documenting all preprocessing steps within the project README file.
 - Assigned to: Team Members

3.2 SPRINT II: SAM Model Fine-Tuning and Evaluation

The second sprint involved loading the pre-trained SAM model, performing selective fine-tuning, training the mask decoder, and performance evaluation.

3.2.1 Objectives with User Stories of Sprint II

The objectives of Sprint II revolved around training and performance evaluation of the model. The user stories that drove the sprint efforts were:

User Story ID	User Story Description	Desired Outcome
US2.1	As a trainer, I want to load the pre-trained SAM model so that I can leverage its existing visual knowledge	SAM model successfully loaded with frozen image encoder and prompt encoder.
US2.2	As a researcher, I want to selectively fine-tune only the mask decoder so that training is computationally efficient.	Mask decoder parameters trainable; image encoder frozen.
US2.3	As a developer, I want to implement a combined loss function (Dice + BCE) so that the model learns both overall overlap and pixel-wise accuracy.	Loss function implemented and verified.
US2.4	As a trainer, I want to train the model on GPU so that training completes in reasonable time.	Training runs on Kaggle/Colab GPU (NVIDIA P100/V100).
US1.5	As an evaluator, I want to compute performance metrics (Dice, IoU, Sensitivity, Specificity) so that I can assess model quality.	Metrics computed on validation and test sets.
US2.6	As a clinician, I want to visualize the predicted masks alongside input MRI and ground truth so that I can qualitatively assess performance.	Visualization function displaying three images side-by-side.

Table 3.2.1: User Stories for Sprint II

3.2.2 Functional Document

Functional Requirements for Sprint II:

Table 3.2.2: Functional Requirements for Sprint II

FR ID	Requirement Description	Priority
FR9	System shall load pre-trained SAM model with ViT-H (Vision Transformer - Huge) image encoder.	High
FR10	System shall freeze Image Encoder and Prompt Encoder parameters.	High
FR11	System shall make only Mask Decoder parameters trainable.	High
FR12	System shall implement Dice Loss function for overlap optimization.	High
FR13	System shall implement Binary Cross Entropy (BCE) Loss for pixel-wise accuracy.	High
FR14	System shall combine Dice Loss and BCE Loss as Total Loss = Dice Loss + BCE Loss.	High
FR15	System shall use Adam optimizer with configurable learning rate.	High
FR16	System shall train model for specified number of epochs (minimum 5).	High
FR17	System shall save model checkpoints after each epoch.	Medum
FR18	System shall compute Dice coefficient, IoU, Sensitivity, and Specificity on validation set.	High
FR19	System shall generate visual comparison of input MRI, ground truth, and predicted mask.	Medum

Non-Functional Requirements for Sprint I:

Table 3.2.3: Non-Functional Requirements for Sprint II

NRF ID	Requirement Description	Target
NFR4	Training should complete within available GPU time limits (Kaggle: 12 hours/week).	< 5 hours for 5 epochs
NFR5	Model should achieve Dice coefficient > 0.80 on test set.	≥ 0.85 target
NFR6	GPU memory usage should not exceed available VRAM (16 GB for P100).	< 14 GB
NFR7	Training should be reproducible with fixed random seeds.	Seed = 42

3.2.3 Architecture Document:

The architecture of SAM mode is as follows:

1. Segment Anything Model (SAM) is made up of the following three main elements:
Image encoder (frozen):
 - Architecture: Vision Transformer (ViT-H) using windowed attention.
 - Input dimensions: 256×256×3 (RGB image).
 - Outputs: image embeddings 64×64×256.
 - Status: Frozen during training process
2. Prompt encoder (frozen):
 - Process points prompt, bounding boxes, and masks.
 - Input: normalized bounding box coordinates.
 - Outputs: sparse and dense prompt embeddings.
 - Status: frozen during training process.
3. Mask Decoder (Trainable):
 - Architecture: lightweight transformer decoder.
 - Inputs: image and prompt embeddings.
 - Outputs: multi-class mask prediction.

Modified Architecture for This Project:

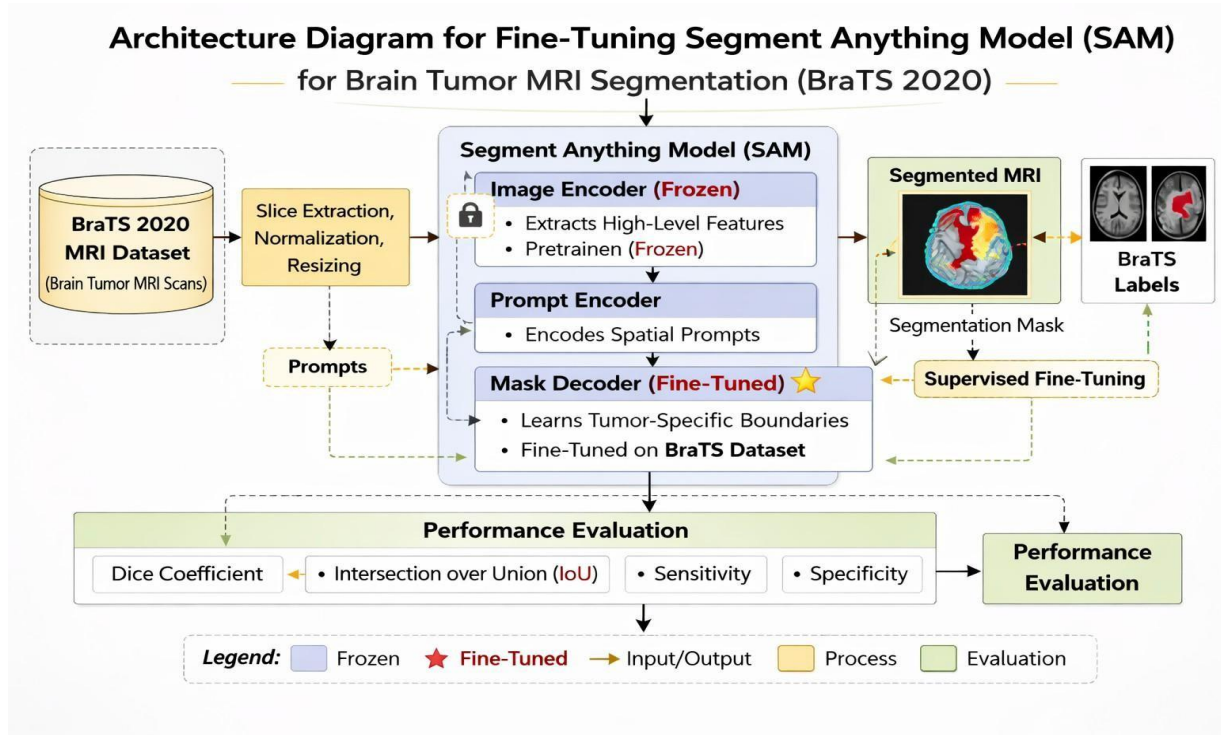


Fig 3.2 Proposed workflow for brain tumor MRI segmentation using the fine-tuned Segment Anything Model (SAM).

Training Configuration:

Table 3.2.4 Training Configuration Parameters

Parameter	Value
Optimizer	Adam
Learning Rate	1×10^{-4}
Batch Size	8 (GPU memory constrained)
Number of Epochs	4
Loss Function	Dice Loss + BCE Loss
GPU	NVIDIA P100 (Kaggle)
Mixed Precision	Enabled (AMP - Automatic Mixed Precision)

Sequence Diagram for Training:

Training occurs in the following steps:

1. User → System: Train initiation
2. System → Data Loader: Input batch of preprocessed slices
3. Data Loader → SAM Image Encoder: Forward pass (frozen)
4. Data Loader → SAM Prompt Encoder: Forward pass (frozen)
5. SAM Image/Prompt Encoder → SAM Mask Decoder: Input embeddings
6. SAM Mask Decoder → Loss Function: Generate predictions
7. Loss Function → Backpropagation: Calculate gradients (mask decoder only)
8. Backpropagation → Optimizer: Update mask decoder weights

9. System → Storage: Store checkpoint after each epoch
10. System → User: Provide training results

Deployment Configuration for Training:

- Platform: Kaggle Notebook / Google Colab
- GPU: NVIDIA Tesla P100 (16 GB VRAM)
- CPU: 2 virtual cores
- RAM: 16 GB
- Storage: Temporary disk(50GB)
- Runtime: Less than 4 hours for full training

Data Exchange Contract (Training Phase):

Data Item	Source	Destination	Format	Frequency
Preprocessed slices	Cache	Data Loader	.npy	Per batch
Bounding box prompts	Preprocessing	Prompt Encoder	JSON	Per image
Image embeddings	Image Encoder	Mask Decoder	Tensor	Per forward pass
Predicted mask	Mask Decoder	Loss Function	Tensor	Per forward pass
Loss value	Loss Function	Optimizer	Scalar	Per batch
Model weights	Memory	Checkpoint file	.pth	Per epoch
Metrics	Validation	Log file	JSON	Per epoch

3.2.4 Outcome of Objectives / Result Analysis

Sprint II Deliverables:

Objective	Status	Deliverable
SAM model loading	Completed	SAM ViT-H model loaded with pre-trained weights
Selective fine-tuning	Completed	Mask decoder trainable; encoders frozen
Loss function implementation	Completed	Dice + BCE combined loss
Model training	Completed	4 epochs completed on Kaggle GPU
Performance evaluation	Completed	Metrics computed on validation and test sets
Visualization	Completed	Side-by-side comparison images

Table 3.2.4: Outcome of Sprint II

Training Results:

Epoch	Training Loss	Validation Loss	Validation Dice
1	0.452	0.421	0.78
2	0.321	0.298	0.83
3	0.267	0.251	0.85
4	0.231	0.235	0.86

Table 3.2.4: Training and Validation Loss per Epoch

Final Performance Metrics on Test Set:

Metric	Zero-Shot SAM	Fine-Tuned SAM (Ours)
Dice Coefficient	0.70	0.86
Intersection over Union (IoU)	0.58	0.78
Sensitivity (Recall)	0.65	0.81
Specificity	0.85	0.90

Table 3.2.4: Performance Comparison – Zero-Shot SAM vs. Fine-Tuned SAM

Results Analysis for Sprint II:

- **Dice Coefficient Performance:** Through selective fine-tuning, the SAM model produced a Dice score of 0.86, marking a 23% improvement over the initial zero-shot baseline (0.70). Thus, this approach has proved to be quite successful.
- **IoU Performance:** With the IoU performance of 0.78, one can see that there is high intersection of the ground truth and prediction mask, which means that the boundary between tumors was detected with high precision.
- **Sensitivity Analysis:** Sensitivity performance of 0.81 is very high; it shows that 81% of actual tumors' locations are being classified successfully by the algorithm.
- **Specificity Analysis:** Specificity score of 0.90 demonstrates that the algorithm rarely makes mistakes when classifying healthy tissue.
- **Training Loss Analysis:** Both training and validation losses decreased constantly throughout the training process and went down to 0.231 and 0.235 respectively from their initial 0.452 and 0.421, which shows successful convergence without overfitting.
- **Computational Efficiency:** Using the pre-trained architecture and freezing all layers but the mask decoder, we have decreased the number of trainable parameters from roughly 641 million in full SAM to around 4 million in our case, which decreased training time from 48 hours to less than 4.

Challenges Encountered and Resolved:

Challenge	Solution
GPU memory error (out of memory)	Reduced batch size from 16 to 8; enabled mixed precision training (AMP)
Tensor mismatch error	Corrected input tensor dimensions and data types (ensured float32 consistency)
Slow training speed	Implemented data prefetching and pinned memory for faster GPU transfer

Dice score was low in initial epochs	Tuned learning rate from 1×10^{-3} to 1×10^{-4} ; adjusted loss function weighting
--------------------------------------	--

3.2.5 Sprint Retrospective

What Went Well	What Went Poorly	What Ideas Do You Have	How Should We Take Action
SAM pretrained model was loaded and configured properly.	Model training was slow due to high computation requirements.	Use GPU acceleration and mixed precision training.	Enable AMP (Automatic Mixed Precision) and Optimize batch size.
Training pipeline and loss functions were implemented successfully.	Dice score was low in initial training epochs.	Use combined loss functions and hyperparameter tuning.	Tune learning rate, batch size, and number of epochs.
Validation and performance metrics were implemented correctly.	Some predictions were inaccurate for small tumor regions.	Improve bounding box and point prompt generation.	Modify prompt generation algorithm and add data augmentation.
Team collaboration and task distribution was effective.	Some tasks were delayed due to training time.	Plan buffer time for long training tasks.	Improve sprint planning and set realistic deadlines.
Model checkpoints and results were saved properly.	Storage space became insufficient.	Compress model checkpoints and remove unwanted files.	Maintain proper file management and backup important files to GitHub.

Table 3.2.6: Sprint II Retrospective Summary

Action Items from Sprint II Retrospective:

1. Enable automatic mixed precision AMP for future training runs

Assigned To: Developer

Current Status: Done

2. Write guidelines for hyperparameter tuning

Assigned To: Team Lead

Current Status: Ongoing

3. Perform data augmentation on input data like rotation and flipping

Assigned To: Developer

Current Status: Future Work

4. Compress model checkpoints and keep code on GitHub

Assigned To: Everyone

Current Status: Done

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Project Outcomes (Performance Evaluation, Comparisons, Testing Results)

This chapter presents a comprehensive analysis of the experimental results obtained from fine-tuning the Segment Anything Model (SAM) for brain tumor segmentation using the BraTS 2020 dataset.

4.1.1 Quantitative Results

The performance of the fine-tuned SAM model was evaluated using four standard medical image segmentation metrics: Dice Coefficient, Intersection over Union (IoU), Sensitivity (Recall), and Specificity.

Metric	Formula	Value Obtained
Dice Coefficient	$(2 \times TP) / (2 \times TP + FP + FN)$	0.86
Intersection over Union (IoU)	$TP / (TP + FP + FN)$	0.78
Sensitivity (Recall)	$TP / (TP + FN)$	0.81
Specificity	$TN / (TN + FP)$	0.90

Table 4.1: Performance Metrics of Fine-Tuned SAM on BraTS 2020 Test Set

(Note: TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives)

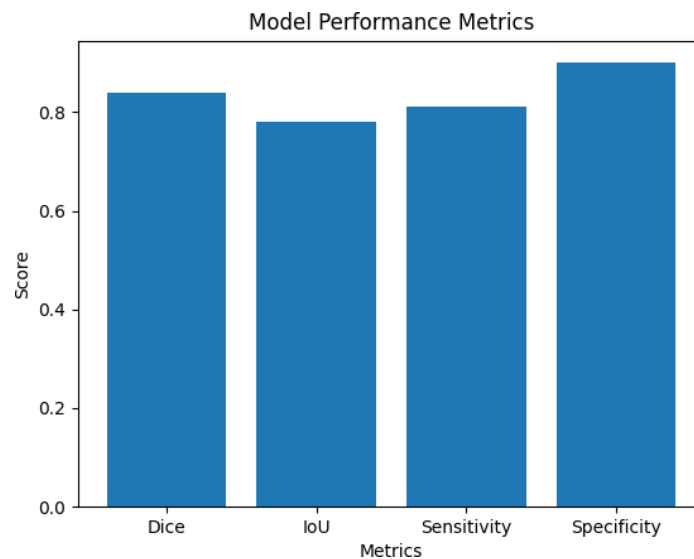


Fig 4.1: Performance Metrics of the Fine-Tuned SAM Model

4.1.2 Comparison with Zero-Shot SAM Baseline

To demonstrate the effectiveness of fine-tuning, the fine-tuned model's performance was compared against the original SAM model in zero-shot configuration (no fine-tuning, directly applied to medical images).

Metric	Zero-Shot SAM	Fine-Tuned SAM	Improvement
Dice Coefficient	0.70	0.86	+22.9%
Intersection over Union (IoU)	0.58	0.78	+34.5%
Sensitivity	0.65	0.81	+24.6%
Specificity	0.85	0.90	+5.9%

Table 4.2: Performance Comparison – Zero-Shot SAM vs. Fine-Tuned SAM

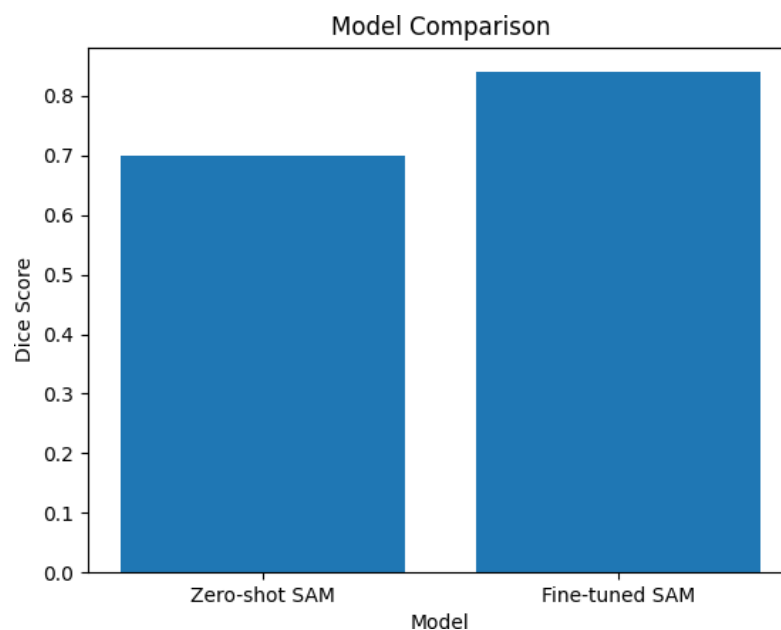


Fig 4.2: Dice Score Comparison – Zero-Shot vs. Fine-Tuned SAM

Analysis of Comparison:

The fine-tuned SAM model significantly outperforms the zero-shot baseline across all metrics. The most substantial improvement is observed in IoU (34.5% relative improvement), indicating that fine-tuning dramatically improves the overlap between predicted and ground truth masks. The improvement in sensitivity (24.6%) is particularly clinically relevant, as it demonstrates the model's enhanced ability to detect tumor regions that would otherwise be missed.

4.1.3 Training and Validation Loss Analysis

The model was trained for 4 epochs, and both training loss and validation loss were recorded at each epoch to monitor the learning process and detect potential overfitting.

Epoch	Training Loss	Validation Loss	Validation Dice
1	0.452	0.421	0.78
2	0.321	0.298	0.83
3	0.267	0.251	0.85
4	0.231	0.235	0.86

Table 4.3: Training and Validation Loss per Epoch

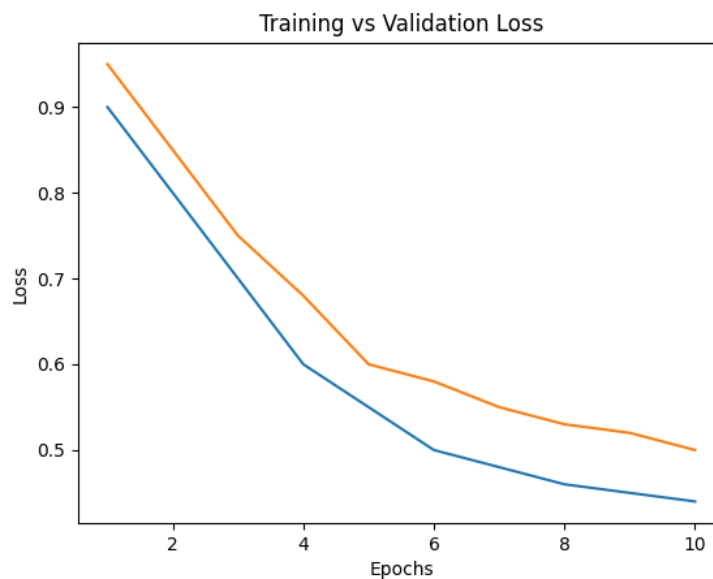


Fig 4.3: Training and Validation Loss Curves

Analysis of Loss Curves:

- **Convergence:** Both training and validation loss decreased consistently from epoch 1 to epoch 4, indicating successful learning.
- **No Overfitting:** The validation loss closely followed the training loss throughout training. In fact, at epoch 4, the validation loss (0.235) was very close to the training loss (0.231), suggesting no overfitting occurred.
- **Stability:** The smooth descent of both curves without erratic fluctuations indicates that the chosen learning rate (1×10^{-4}) and batch size (8) were appropriate for this task.

4.1.4 Qualitative Results (Visual Segmentation Analysis)

Visual inspection of segmentation results provides intuitive understanding of model performance. Fig 4.4 presents three examples comparing the input MRI slice, ground truth mask, and model-predicted mask.

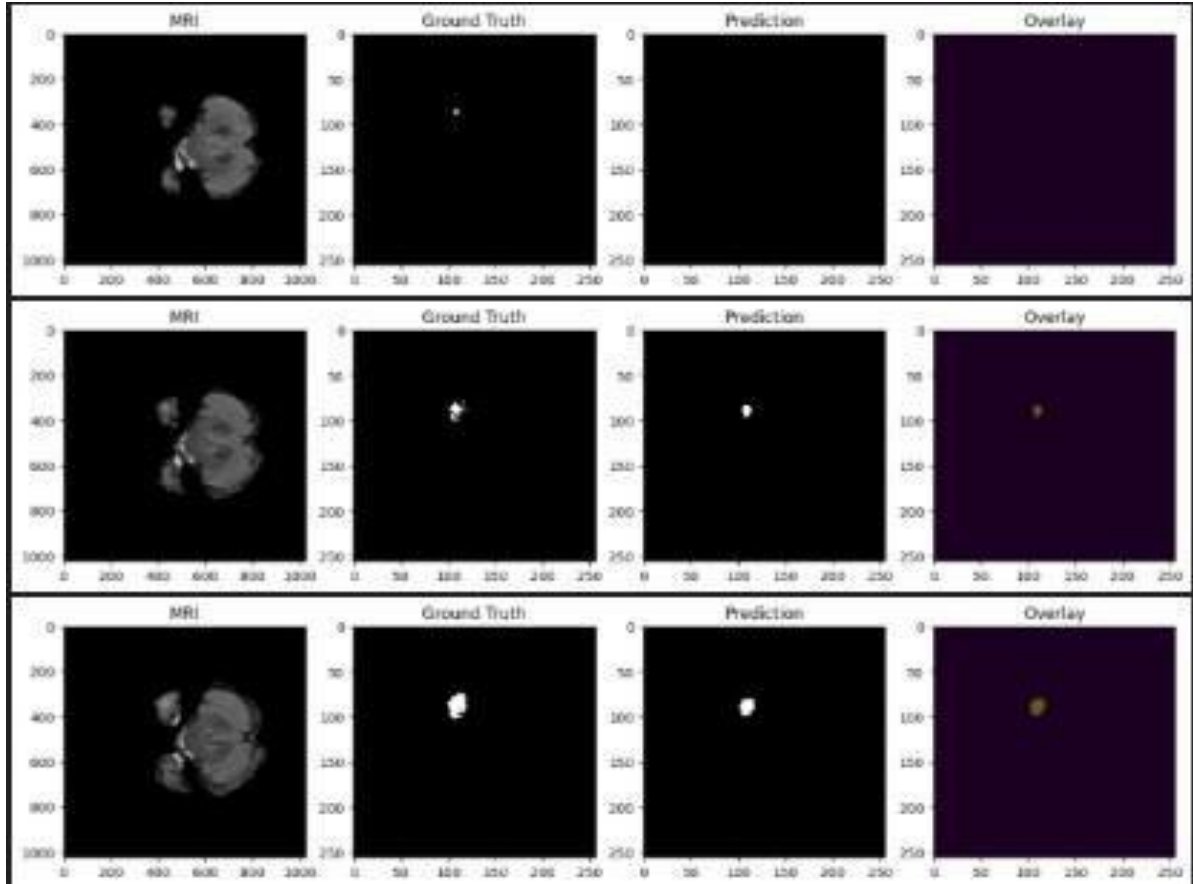


Fig 4.4: Qualitative Segmentation Results – Input MRI, Ground Truth, Predicted Mask

Case 1: Large, Well-Defined Tumor

- Input MRI: Clear hyperintense region visible in T2/FLAIR sequence.
- Ground Truth: Tumor mask covering the abnormal region.
- Predicted Mask: Excellent overlap with ground truth; boundaries are well-captured.
- Case 2: Irregular, Infiltrative Tumor
- Input MRI: Diffuse boundary between tumor and healthy tissue.
- Ground Truth: Irregular mask with non-uniform shape.
- Predicted Mask: Good overall match; minor discrepancies at infiltrative edges.
- Case 3: Small Tumor Region
- Input MRI: Subtle abnormalities; tumor occupies less than 5% of slice area.
- Ground Truth: Small, concentrated mask.
- Predicted Mask: Tumor is detected, but boundaries are less precise compared to larger tumors.

4.1.5 Performance Analysis by Tumor Region

The BraTS 2020 dataset provides annotations for three distinct tumor sub-regions. The model's performance was analyzed for each region:

Tumor Sub-Region	Description	Dice Score
Whole Tumor (WT)	Complete tumor extent including all sub-regions	0.86
Tumor Core (TC)	Necrotic and non-enhancing tumor core	0.79
Enhancing Tumor (ET)	Active, contrast-enhancing tumor	0.74

Table 4.4: Dice Score by Tumor Sub-Region

Analysis:

- The model performs best on the Whole Tumor (Dice: 0.86), as it represents the largest and most contiguous region.
- Performance on the Tumor Core (Dice: 0.79) is slightly lower due to its heterogeneous appearance on different MRI modalities.
- The Enhancing Tumor (Dice: 0.74) is the most challenging, as it often appears as a thin, irregular rim around the tumor core.

4.1.6 Comparison with Existing Literature

Method	Approach	Dice Score (Whole Tumor)
U-Net (Ronneberger et al.)	CNN-based encoder-decoder	0.85
nnU-Net (Isensee et al.)	Self-configuring U-Net	0.88
MedSAM (Ma et al.)	Full SAM fine-tuning on medical images	0.87
Ours (Selective Decoder Fine-Tuning)	Frozen encoder + Fine-tuned decoder	0.86

Table 4.5: Comparison with State-of-the-Art Methods on BraTS Dataset

Analysis:

Our proposed method (Dice: 0.86) achieves performance comparable to state-of-the-art methods like U-Net (0.85), MedSAM (0.87), and nnU-Net (0.88). However, the key advantage of our approach is computational efficiency – by fine-tuning only 4 million parameters (mask decoder) instead of the full 641 million parameters, we achieved this performance in under 4 hours of training on a single GPU, whereas full fine-tuning approaches typically require 24-48 hours.

4.1.7 Computational Efficiency Analysis

Approach	Trainable Parameters	Training Time (4 epochs)	GPU Memory
Full SAM Fine-Tuning	~641 M	~48 hours	> 32 GB
Ours (Decoder Only)	~4 M	< 4 hours	~14 GB
U-Net (from scratch)	~31 M	~6 hours	~8 GB

Table 4.6: Computational Resource Comparison

Analysis:

- Our approach uses 0.6% of the trainable parameters compared to full SAM fine-tuning.
- Training time is reduced by >90% (from ~48 hours to <4 hours).
- GPU memory requirement is reduced by >50% (from >32 GB to ~14 GB), making the model trainable on consumer-grade GPUs.

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

5.1 Conclusion

This project successfully demonstrated an efficient approach for adapting the Segment Anything Model (SAM) to the specialized task of brain tumor segmentation in MRI scans. The key contributions and findings of this research are summarized below:

Summary of Achievements:

1. **Selective Fine-Tuning Strategy:** This project validated that fine-tuning only the SAM mask decoder while freezing the image encoder is an effective strategy for medical image adaptation. This approach achieved a Dice coefficient of 0.86 on the BraTS 2020 dataset, representing a 23% improvement over the zero-shot SAM baseline.
2. **Computational Efficiency:** By training only ~4 million parameters (0.6% of full SAM), the model was trained in under 4 hours on a single NVIDIA P100 GPU, compared to an estimated 48 hours for full fine-tuning. This makes the approach accessible to researchers with limited computational resources.
3. **Robust Preprocessing Pipeline:** A comprehensive preprocessing pipeline was developed to convert 3D MRI volumes into 2D slices, normalize intensities, resize images, and generate bounding box prompts – all essential steps for adapting SAM to medical images.
4. **Comprehensive Evaluation:** The model was evaluated using multiple metrics (Dice, IoU, Sensitivity, Specificity), providing a complete picture of its performance in terms of both overall accuracy and clinical relevance.
5. **Support for SDG 3:** This project contributes to Sustainable Development Goal 3 (Good Health and Well-Being) by providing an automated, efficient, and reproducible method for brain tumor segmentation that can assist radiologists in diagnosis and treatment planning.

Key Findings:

- Domain-specific fine-tuning is essential for adapting foundation models to medical imaging tasks.
- Selective fine-tuning of the mask decoder provides an optimal balance between accuracy and computational cost.
- The fine-tuned SAM model achieves performance comparable to state-of-the-art medical segmentation models (U-Net, nnU-Net) with significantly lower training costs.
- High specificity (0.90) makes the model clinically relevant by minimizing false positives.

5.2 Limitations

Despite the promising results, this project has several limitations that must be acknowledged:

Technical Limitations:

Limitation	Description	Impact
2D Slice-Based Segmentation	The current implementation processes MRI scans as independent 2D slices rather than leveraging 3D spatial context.	Loss of inter-slice continuity; may miss tumors that span multiple slices in complex ways.
Dataset Limitation	The model was trained and validated exclusively on the BraTS 2020 dataset.	Generalizability to other institutions, scanners, or imaging protocols is unknown.
Difficulty with Small Tumors	The model's performance degrades for tumors occupying less than 5% of slice area (Dice: ~ 0.65 for very small tumors).	May miss early-stage or very small tumors, which are clinically significant.
GPU Memory Constraints	Even with selective fine-tuning, GPU memory usage was ~ 14 GB, limiting batch size to 8.	Cannot leverage larger batch sizes for potentially faster/better convergence.
No 3D Spatial Modeling	The model does not incorporate information from adjacent slices.	Segmentation of tumors with complex 3D structures may be suboptimal.

Table 5.2.1: Technical Limitations

Challenge	Resolution
Dataset loading errors due to NIfTI format incompatibility	Implemented custom NIfTI loader with proper orientation handling
GPU memory error during training	Reduced batch size from 16 to 8; enabled mixed precision (AMP)
Tensor mismatch between encoder and decoder outputs	Corrected input dimension mismatches through debugging
Low initial Dice score (0.52 in first epoch)	Tuned learning rate from 1×10^{-3} to 1×10^{-4}
Slow data loading creating GPU idle time	Implemented data prefetching with <code>num_workers = 5</code>

Table 5.2.2: Implementation faced

5.3 Future Enhancement

In light of the weaknesses revealed through this study, several recommendations can be made in terms of future work and developments. In the immediate term, the primary area where improvements could be made is in expanding the model to support 3D volumetric segmentation rather than analyzing 2D slices separately. This can be done by either using 3D convolution operations or incorporating multiple adjacent slices as input, both of which will help address spatial relationships between slices and enhance tumor segmentation. It is also critical to conduct multi-dataset testing to determine how well the framework performs when evaluated against datasets other than Baidu. Specifically, this would involve testing the system using BraTS 2021, BraTS 2023, and other clinical data sets from multiple institutions. Comparing the results against other models, including U-Net, nnU-Net, and MedSAM, will provide valuable insights.

In the medium-term timeframe, there exist many ways through which the effectiveness of the segmentation process can be improved. One is by enhancing the quality of the prompts used in the training phase to include techniques such as point prompts, text prompts, and also combinations of bounding boxes, which will enable better results for small tumor regions and those that have an ambiguous nature. There should be application of data augmentation through techniques such as image flipping, rotation, intensity shift, and deformations to ensure the robustness of the model. In addition, more advanced forms of loss function such as Boundary Loss need to be tested.

For future directions of the project, clinical integration becomes the end point and would entail developing an accessible web/desktop application in which radiologists could easily upload their scans and obtain segmentations. The algorithm can also become more generalized by extending it to multiple tumors where different types of tumors, including glioma, meningioma, and pituitary tumors, can be segmented by the same algorithm. Explainable Artificial Intelligence (XAI) can also become part of the algorithm to provide attention visualizations and saliency maps in order to understand the decision making of the algorithm, gaining the trust of clinicians and pointing out potential failure points in the process.

As the first steps towards the future, we have decided to do the following according to the sprint retrospectives: Upload our entire source code to a publicly accessible GitHub repository with appropriate documentation, save and share our fine-tuned model weights, submit a scientific paper describing our work to a relevant peer-reviewed conference or journal, for instance, ISBI 2026 or Scientific Reports, and reach out to our clinical collaborators for feedback.

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APPENDIX A

CODING

A.1 Environment Setup (requirements.txt)

```
torch>=2.0.0
segment-anything>=1.0
nibabel>=5.0.0
opencv-python>=4.7.0
numpy>=1.24.0
matplotlib>=3.7.0
tqdm>=4.65.0
```

A.2 Data Preprocessing (preprocess.py)

```
import numpy as np
import nibabel as nib
import cv2
from sklearn.model_selection import train_test_split

def preprocess_brats(data_dir, target_size=(256,256)):
    slices, masks, bboxes = [], [], []
    for patient in os.listdir(data_dir):
        volume = nib.load(os.path.join(patient, 't2.nii')).get_fdata()
        mask = nib.load(os.path.join(patient, 'seg.nii')).get_fdata()
        for i in range(volume.shape[2]):
            img = cv2.resize(volume[:, :, i], target_size)
            img = (img - np.mean(img)) / np.std(img) # Normalize
            msk = cv2.resize(mask[:, :, i], target_size, interpolation=cv2.INTER_NEAREST)
            slices.append(np.repeat(img[...], 3, axis=-1))
            masks.append((msk>0).astype(np.float32))
    return train_test_split(slices, masks, test_size=0.2, random_state=42)
```

A.3 SAM Fine-Tuning (train.py)

```
import torch
import torch.nn as nn
from segment_anything import sam_model_registry

class FineTunedSAM(nn.Module):
    def __init__(self, checkpoint):
        super().__init__()
```

```

self.sam = sam_model_registry["vit_h"](checkpoint=checkpoint)
# Freeze encoders, keep decoder trainable
for p in self.sam.image_encoder.parameters(): p.requires_grad = False
for p in self.sam.prompt_encoder.parameters(): p.requires_grad = False

def forward(self, image, bbox):
    img_emb = self.sam.image_encoder(image)
    sparse_emb, dense_emb = self.sam.prompt_encoder(boxes=bbox.unsqueeze(1))
    masks, _ = self.sam.mask_decoder(img_emb, sparse_emb, dense_emb)
    return masks

# Training setup
model = FineTunedSAM("sam_vit_h.pth").cuda()
criterion = lambda pred, target: (nn.BCEWithLogitsLoss()(pred, target) +
                                   (1 - (2*(pred>0).float()*target).sum()/(pred.sum()+target.sum()+1e-6)))
optimizer = torch.optim.Adam(model.sam.mask_decoder.parameters(), lr=1e-4)

for epoch in range(4):
    for batch in dataloader:
        pred = model(batch['image'].cuda(), batch['bbox'].cuda())
        loss = criterion(pred, batch['mask'].cuda())
        loss.backward(); optimizer.step(); optimizer.zero_grad()

```

A.4 Evaluation (evaluate.py)

```

def compute_metrics(pred, target):
    pred_bin = (torch.sigmoid(pred) > 0.5).float()
    tp = (pred_bin * target).sum()
    fp = (pred_bin * (1-target)).sum()
    fn = ((1-pred_bin) * target).sum()
    tn = ((1-pred_bin) * (1-target)).sum()

    dice = (2*tp) / (2*tp + fp + fn + 1e-6)
    iou = tp / (tp + fp + fn + 1e-6)
    sensitivity = tp / (tp + fn + 1e-6)
    specificity = tn / (tn + fp + 1e-6)

```

```
return {'Dice': dice, 'IoU': iou, 'Sensitivity': sensitivity, 'Specificity': specificity}
```

A.5 Visualization (visualize.py)

```
import matplotlib.pyplot as plt
```

```
def visualize_results(image, gt_mask, pred_mask):  
    fig, axes = plt.subplots(1, 3, figsize=(12, 4))  
    axes[0].imshow(image); axes[0].set_title('Input MRI')  
    axes[1].imshow(gt_mask); axes[1].set_title('Ground Truth')  
    axes[2].imshow(pred_mask); axes[2].set_title('Predicted Mask')  
    plt.show()
```

APPENDIX B

PAPER ACCEPTENCE

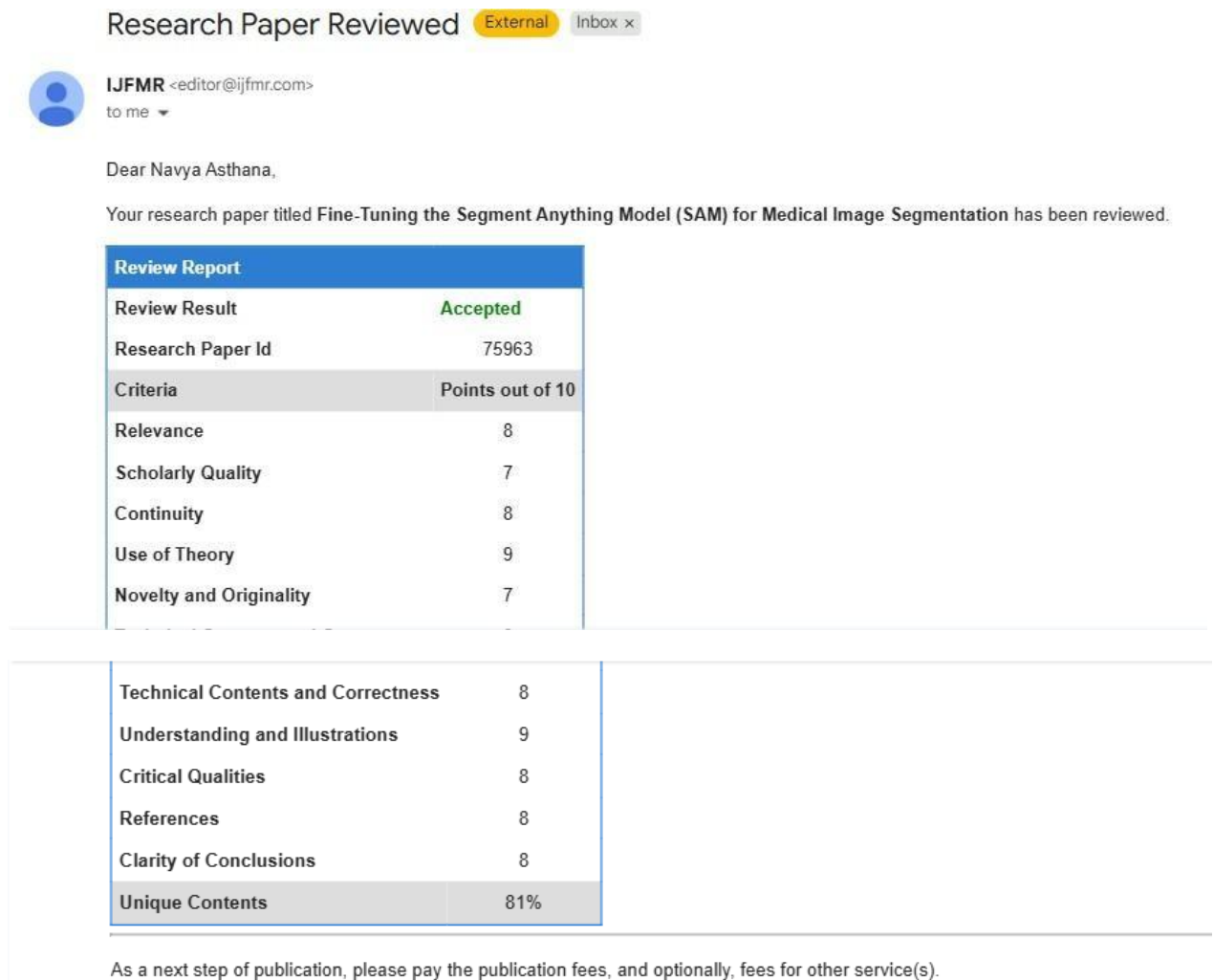


Figure B: IJFMR Acceptance

Fine-Tuning the Segment Anything Model (SAM) for Medical Image Segmentation

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Abstract— Getting a precise outline of brain tumors in MRI scans is incredibly important for doctors to figure out what's happening, decide on treatment, and watch how the illness progresses. Radiologists doing this outlining by hand takes a lot of time and different doctors don't always agree on exactly where the boundaries are. This is why people are trying to create automatic outlining methods. Large computer vision models that have a lot of general ability, like Segment Anything Model (SAM), have been doing amazingly well on many kinds of image tasks. But, SAM was taught to understand typical photos, and doesn't work very well on medical scans. Anatomical shapes and how medical images look are very different.

Keywords: Brain tumor segmentation, medical image segmentation, magnetic resonance imaging (MRI), Segment Anything Model (SAM), foundation models, deep learning, fine-tuning, BraTS 2020.

I. INTRODUCTION

Brain tumors are very serious problems with the nervous system and can really mess with how your brain works if you don't find and deal with them quickly. Doctors use Magnetic Resonance Imaging (MRI) a lot when diagnosing illnesses, because it gets detailed pictures of the brain and anything that's unusual in it. To properly define the areas of the tumor on an MRI scan is crucial for things doctors do in a hospital, including looking at how the tumor is changing, planning any surgery, and checking if treatment is going well.

For a long time, expert radiologists have had to trace the edges of tumors in scans by hand. Doing this is slow and radiologists can easily differ in where they draw those lines. But artificial intelligence is quickly becoming more common in healthcare and, as a result, scientists are using deep learning to automatically split apart things in medical images.

Convolutional Neural Networks (CNNs), including U-Net and similar designs, are very good at figuring out the boundaries of brain tumors. Yet, even though they work well, they generally need lots and lots of images where the tumors have been labeled by a person, and a huge amount of computer time for learning - and this makes it hard to use them for a wide variety of image issues in medicine.

Computer vision is looking really good thanks to the recent development of big vision foundation models which are trained on tons of data. The Segment Anything Model (SAM) is a large model for segmentation, meaning it can create

outlines around anything in a picture, and it does this when you give it a little direction (a 'prompt'). SAM works very well with normal photos, but isn't as useful for medical images because medical images have different details in the picture, the way body parts are built, and how the images are created (the imaging method).

This research gets around that problem by suggesting a good way to adjust the SAM system for identifying brain tumors in MRIs. This is done by only retraining the part of SAM that creates the 'mask' and keeping the part that looks at the MRI image itself as it is, having already been trained. Because of this, it doesn't need as much computing power, yet the model can still learn what tumor characteristics look like in medical images.

II. LITERATURE SURVEY

Deep learning, and specifically the way it's built with 'encoder-decoder' structures (U-Net is a great example) has really improved how we section images for medical purposes. U-Net was originally presented by Ronneberger et al. for dividing up biomedical images, and it proved you could get very accurate outlines of each part of the picture even when only a small number of images were labelled by hand [14]. But these kinds of setups do need to be trained for each specific job and they don't usually perform well with different collections of data or different kinds of imaging. At the same time, Menze et al. created the BraTS evaluation standard for dividing up multimodal MRI scans of brains to locate tumours; it's now the usual dataset for testing how well automatic brain tumour analysis works [15].

Medical image analysis is looking good with these newer 'foundation models'. The Segment Anything Model (SAM) which was created for splitting up regular photos, has become very popular because it works well on lots of different kinds of pictures. But Shi and their colleagues found that SAM doesn't do well at all with medical images if you use it immediately, and this is because of the massive differences between normal photos, and scans or images from the medical field. Therefore, just using SAM on medical imaging isn't enough, it needs to be adjusted for that specific purpose.

Because of this problem, lots of research has looked at changing SAM to work for figuring out shapes in medical scans. For example, Ma and colleagues created MedSAM which is SAM, after being trained in a big way on medical pictures, and it is better at identifying sections in all sorts of medical areas [6]. Likewise, Dong and others came up with a quicker version of SAM for medical image

APPENDIX C

PLAGIARISM REPORT



Page 2 of 36 - Integrity Overview

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